

## CHILLICOTHE, MO, USA

WMO: 724464

Lat: 39.823N

Lon: 93.579W

Elev: 234

StdP: 98.54

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 53916

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.6      | -14.4 | -22.3                         | 0.5 | -16.8 | -20.0 | 0.7 | -13.4 | 8.9                   | -1.0 | 8.1 | 0.9  | 1.8                   | 310  | 0.450 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.0                   | 35.0            | 24.9 | 33.4 | 24.6 | 32.1 | 24.1 | 26.7                | 32.5 | 25.8 | 31.3 | 25.0 | 30.2 | 3.5                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.1                            | 20.8 | 30.7 | 24.1 | 19.5 | 29.5 | 23.3 | 18.6 | 28.4 | 84.9          | 32.2 | 80.9 | 31.3 | 77.0 | 30.1 | 29.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.0               | 7.1  | 6.0 | -22.3                      | 37.1 | 3.0                | 1.9 | -24.4                                              | 38.4 | -26.2      | 39.5 | -27.8      | 40.6 | -30.0      | 41.9 |    |    |
|                   |      |     | -22.7                      | 27.9 | 2.8                | 1.1 | -24.7                                              | 28.7 | -26.3      | 29.3 | -27.9      | 29.9 | -30.0      | 30.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.4   | -2.3  | -0.7  | 6.8   | 12.6  | 18.0  | 23.4  | 25.5  | 24.5  | 20.0  | 13.0  | 6.5   | 0.4  |
|                                                                                  | DBStd       | 10.96  | 6.71  | 6.67  | 6.30  | 5.31  | 4.64  | 3.31  | 3.10  | 3.05  | 4.25  | 5.29  | 5.55  | 5.83 |
|                                                                                  | HDD10.0     | 1317   | 382   | 304   | 139   | 33    | 3     | 0     | 0     | 0     | 0     | 28    | 130   | 298  |
|                                                                                  | HDD18.3     | 2903   | 639   | 533   | 361   | 182   | 64    | 3     | 0     | 1     | 29    | 180   | 354   | 555  |
|                                                                                  | CDD10.0     | 2187   | 1     | 4     | 39    | 111   | 251   | 401   | 481   | 448   | 301   | 122   | 26    | 2    |
|                                                                                  | CDD18.3     | 731    | 0     | 0     | 2     | 10    | 54    | 154   | 223   | 191   | 81    | 15    | 1     | 0    |
|                                                                                  | CDH23.3     | 7326   | 0     | 0     | 16    | 108   | 487   | 1533  | 2447  | 1848  | 746   | 139   | 2     | 0    |
|                                                                                  | CDH26.7     | 2931   | 0     | 0     | 1     | 18    | 131   | 592   | 1117  | 788   | 253   | 33    | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.5    | 2.9   | 3.0   | 3.3   | 3.3   | 2.5   | 2.3   | 1.9   | 1.7   | 1.9   | 2.3   | 2.7   | 2.7  |
| Precipitation                                                                    | PrecAvg     | 1002   | 31    | 47    | 67    | 98    | 134   | 127   | 115   | 109   | 107   | 68    | 55    | 43   |
|                                                                                  | PrecMax     | 1522   | 73    | 128   | 155   | 155   | 315   | 242   | 400   | 297   | 316   | 194   | 165   | 108  |
|                                                                                  | PrecMin     | 656    | 6     | 7     | 9     | 29    | 27    | 27    | 22    | 0     | 19    | 4     | 5     | 3    |
|                                                                                  | PrecStd     | 212    | 17    | 29    | 37    | 38    | 71    | 59    | 79    | 60    | 71    | 46    | 38    | 27   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 17.2  | 21.3  | 25.3  | 28.9  | 32.1  | 35.1  | 37.8  | 37.5  | 33.6  | 30.0  | 23.1  | 17.6 |
|                                                                                  |             | MCWB   | 11.1  | 12.0  | 17.4  | 19.3  | 22.3  | 24.8  | 23.9  | 23.9  | 24.3  | 22.0  | 15.6  | 13.5 |
|                                                                                  | 2%          | DB     | 13.0  | 16.4  | 22.7  | 26.3  | 29.7  | 33.1  | 35.4  | 34.7  | 31.6  | 27.1  | 20.9  | 13.6 |
|                                                                                  |             | MCWB   | 8.3   | 10.8  | 15.5  | 17.4  | 21.3  | 24.3  | 25.2  | 24.5  | 23.4  | 19.3  | 15.1  | 9.9  |
|                                                                                  | 5%          | DB     | 10.1  | 12.3  | 20.0  | 24.1  | 27.9  | 31.7  | 33.8  | 32.8  | 29.7  | 24.4  | 17.9  | 11.2 |
|                                                                                  |             | MCWB   | 6.5   | 7.9   | 14.0  | 16.5  | 20.3  | 23.7  | 25.1  | 24.4  | 22.4  | 17.9  | 12.6  | 8.1  |
|                                                                                  | 10%         | DB     | 7.1   | 9.0   | 17.3  | 22.1  | 26.3  | 30.1  | 32.2  | 31.0  | 27.6  | 22.1  | 15.5  | 8.7  |
|                                                                                  |             | MCWB   | 4.1   | 5.6   | 12.4  | 15.4  | 19.3  | 22.7  | 24.6  | 23.8  | 20.9  | 16.8  | 11.1  | 5.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.2  | 14.5  | 18.9  | 20.4  | 24.0  | 26.7  | 27.9  | 27.8  | 25.4  | 23.0  | 17.7  | 14.8 |
|                                                                                  |             | MCDB   | 14.8  | 18.5  | 23.9  | 27.3  | 29.7  | 32.5  | 33.5  | 33.8  | 31.5  | 27.8  | 21.4  | 16.8 |
|                                                                                  | 2%          | WB     | 8.9   | 11.1  | 16.9  | 19.1  | 22.6  | 25.5  | 26.9  | 26.5  | 24.3  | 21.4  | 15.5  | 10.9 |
|                                                                                  |             | MCDB   | 11.8  | 15.5  | 20.8  | 24.5  | 28.1  | 31.1  | 32.7  | 32.3  | 29.7  | 25.3  | 19.5  | 12.4 |
|                                                                                  | 5%          | WB     | 6.5   | 8.4   | 15.2  | 17.7  | 21.4  | 24.6  | 26.1  | 25.4  | 23.4  | 19.4  | 13.7  | 8.2  |
|                                                                                  |             | MCDB   | 10.0  | 11.9  | 18.8  | 22.8  | 26.5  | 30.1  | 31.8  | 30.8  | 27.9  | 22.8  | 17.2  | 10.8 |
|                                                                                  | 10%         | WB     | 4.0   | 5.5   | 12.6  | 16.1  | 20.4  | 23.7  | 25.2  | 24.5  | 22.3  | 17.3  | 11.6  | 6.0  |
|                                                                                  |             | MCDB   | 7.0   | 9.0   | 16.9  | 21.1  | 25.0  | 28.9  | 30.4  | 29.3  | 26.2  | 21.5  | 14.8  | 8.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.2  | 10.9  | 11.3  | 11.8  | 11.3  | 10.8  | 11.0  | 11.2  | 12.2  | 11.8  | 11.0  | 9.5  |
|                                                                                  |             | MCDBR  | 14.5  | 16.0  | 14.9  | 14.6  | 12.9  | 12.0  | 12.4  | 13.3  | 13.0  | 14.2  | 14.5  | 13.1 |
|                                                                                  |             | MCWBR  | 10.3  | 10.9  | 8.9   | 8.3   | 6.3   | 5.3   | 4.9   | 5.4   | 5.8   | 7.4   | 9.4   | 9.8  |
|                                                                                  | 5% WB       | MCDBR  | 13.9  | 14.9  | 12.6  | 12.2  | 11.1  | 10.6  | 10.5  | 11.1  | 11.0  | 11.2  | 12.5  | 12.2 |
|                                                                                  |             | MCWBR  | 10.3  | 11.0  | 8.2   | 7.8   | 6.0   | 5.5   | 5.2   | 5.6   | 5.4   | 7.2   | 9.3   | 10.2 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.298  | 0.315 | 0.340 | 0.392 | 0.427 | 0.428 | 0.441 | 0.442 | 0.389 | 0.345 | 0.303 | 0.296 |      |
|                                                                                  | taud        | 2.437  | 2.388 | 2.393 | 2.264 | 2.232 | 2.262 | 2.225 | 2.242 | 2.341 | 2.459 | 2.519 | 2.467 |      |
|                                                                                  | Ebn at Noon | 872    | 904   | 915   | 882   | 855   | 852   | 836   | 826   | 851   | 859   | 863   | 846   |      |
|                                                                                  | Edh at Noon | 80     | 97    | 108   | 131   | 139   | 135   | 139   | 133   | 113   | 90    | 74    | 73    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.03   | 2.89  | 3.77  | 4.86  | 5.68  | 6.34  | 6.36  | 5.56  | 4.76  | 3.28  | 2.26  | 1.77  |      |
|                                                                                  | RadStd      | 0.22   | 0.26  | 0.33  | 0.33  | 0.48  | 0.54  | 0.42  | 0.28  | 0.29  | 0.36  | 0.21  | 0.19  |      |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (33 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page